ROCHESTER INSTITUTE OF TECHNOLOGY · EXTENDED FRESHMAN IMAGING PROJECT

Digital Twins of Artifacts

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In collaboration with the RIT Cary Graphic Arts Collection

RIT College of Science

INTRODUCTION

Preserving fragile history as interactive replicas

Historical artifacts — papyrus manuscripts, cuneiform tablets — are fragile, rare, and often kept out of reach of the people who want to study them.

Digital Twins of Artifacts builds accurate, interactive, web-viewable 3D replicas so these objects can be examined, taught, and displayed without handling the original. It began as RIT's **Freshman Imaging Project**, grew into **Extended-FIP (EFIP)**, and runs with the **RIT Cary Graphic Arts Collection**, which provides the artifacts.

APPROACH

Two purpose-built imaging systems

Each class of artifact is paired with a rig designed for it. Which one an object uses depends on a single question: **does its shape carry meaning in three dimensions?**

- **Flat, detail-rich** objects (papyrus, parchment) → a 2D photometric-stereo rig that recovers how the surface responds to light.
- **Truly three-dimensional** objects (cuneiform tablets, seals) → a 3D photogrammetry rig that reconstructs real geometry.

ACCESS

An interactive web viewer

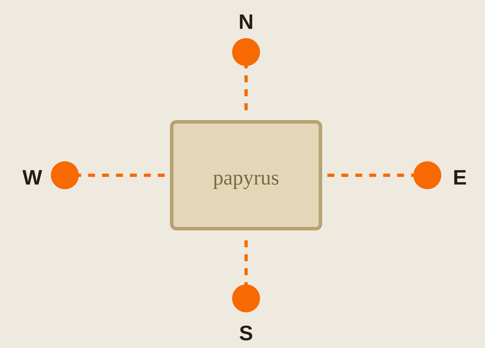
Every finished twin lives in a static three.js gallery that runs in any browser — rotate each artifact, adjust the lighting, and swap the backdrop to study the surface.

- **Hand-tracked magnifier** — a webcam tracks the visitor's fingertip and a magnifying lens follows it across the artifact, like a museum exhibit.
- **Adjustable relighting** reveals ink and relief a fixed view would hide.
- **Runs anywhere** — a static site, shared with a single link.

SYSTEM A · PAPYRUS & FLAT MANUSCRIPTS

2D Photometric-Stereo Rig

A Canon DSLR paired with an Arduino-driven directional lighting array photographs the manuscript once per light direction. Instead of reconstructing geometry, it recovers how the surface *responds to light* — revealing texture, ink, and fine relief.

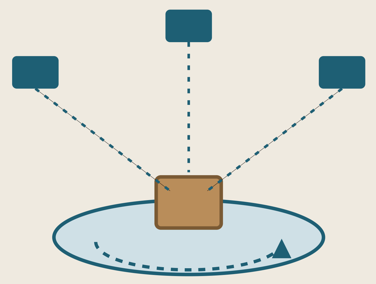


Four directional lights (N/E/S/W) at ~37°. A rotating polarizer captures **cross-polarized** (clean geometry) and **co-polarized** (reveals ink) pairs.

SYSTEM B · CUNEIFORM TABLETS & OBJECTS

3D Photogrammetry Rig

A multi-camera rig on an Arduino-driven turntable photographs the artifact from many angles. GPU-accelerated structure-from-motion and dense stereo build a point cloud and, ultimately, a watertight 3D mesh — a true replica of the object's form.



Multiple cameras shoot the object as the **turntable rotates**. Both sides are captured, then merged into one model.

REFERENCE

Technical specifications

Camera	2D + 3D	Canon DSLRs, tethered; RAW → 16-bit TIFF (~5208 × 3476 px)
Lighting	2D	4 directional lights (N/E/S/W) at θ ≈ 37°; rotating polarizer
Polarization	2D	cross-pol (geometry) + co-pol (ink) capture pairs
Reconstruction	3D	CUDA COLMAP (SfM + MVS) → FPFH/RANSAC + ICP → Open3D Poisson mesh
Rendering	2D + 3D	three.js material baked to GLB; viewer w/ MediaPipe hand tracking

CONCLUSION & FUTURE WORK

A repeatable path to digital preservation

Two complementary rigs and their automated pipelines turn fragile originals into faithful, shareable digital twins — a workflow that repeats across artifact types.

- Expand the collection of scanned artifacts with the Cary Collection.
- Refine capture speed and automation for both rigs.

METHODS

From raw captures to a digital twin

PAPYRUS **Photometric-stereo pipeline** (Python)

0 Segment — ML segmentation isolates the fragment. → **1 Calibrate** — Copy-paper shots remove light falloff. → **2 Core maps** — Normal, albedo, specular, roughness. → **3 Height** — Frankot–Chellappa integration. → **4 Bake** — Maps baked into a three.js GLB.

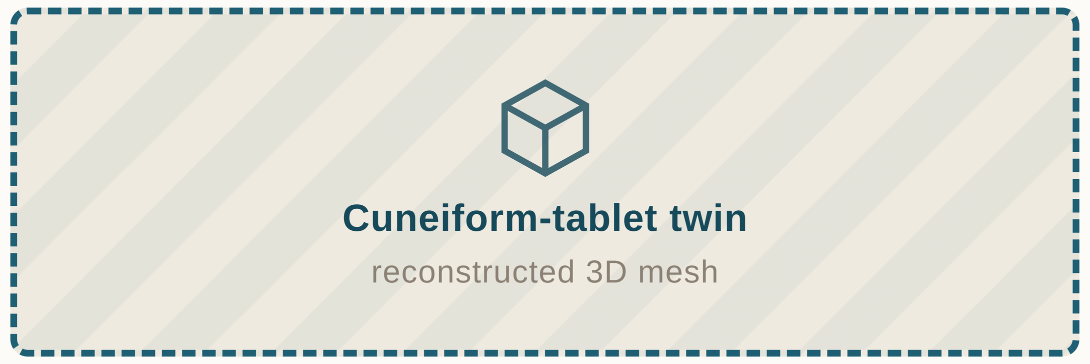
TABLETS **Photogrammetry pipeline** (COLMAP / Open3D)

1 Clean — Background removed from every photo. → **2 Reconstruct** — CUDA COLMAP SfM + dense stereo. → **3 Align** — FPFH + RANSAC + ICP merge both sides. → **4 Mesh** — Open3D Poisson → watertight model.

RESULTS

Outputs

Calibrated surface maps. Building blocks of the papyrus twin, from the polarized-light captures.



Watertight 3D replica. Both faces reconstructed and merged into one model.

Explore in the browser. Rotate, relight, and inspect each twin — no special software.

ACKNOWLEDGEMENTS

Thanks

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Explore the digital twins

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